

AI Mock Interview & Coding Assessment Platform

Complete Backend Development Roadmap

This document contains the complete workflow, architecture, backend roadmap, phase-wise development plan, technologies, database design, advanced concepts, and deployment strategy for building a placement-level backend project using Java Spring Boot.

1. Project Overview

The AI Mock Interview & Coding Assessment Platform is a backend-focused system designed to simulate technical interviews, conduct coding assessments, generate AI-based feedback, and provide analytics for candidates.

Main Goals:

- Build a scalable backend architecture
- Implement secure authentication
- Create real-time interview features
- Support coding evaluations
- Integrate AI APIs
- Demonstrate production-level backend skills

2. Recommended Technology Stack

Backend:

- Java
- Spring Boot
- Spring Security
- Spring Data JPA
- Hibernate
- WebSocket

Database:

- PostgreSQL
- Redis (for caching)

DevOps & Infrastructure:

- Docker
- Docker Compose
- Kafka/RabbitMQ (optional advanced)

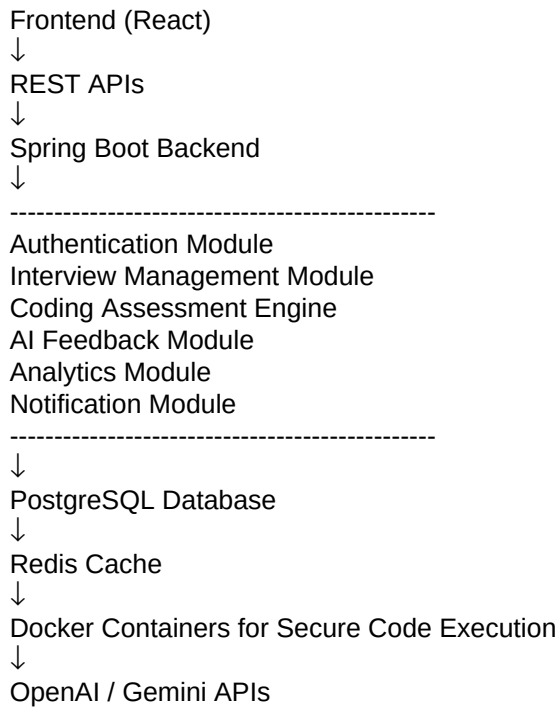
Frontend:

- React (recommended)
- OR
- HTML/CSS/JavaScript

AI Integration:

- OpenAI API
- Gemini API

3. High-Level Architecture



4. Database Design

Important Tables:

1. users

- id
- name
- email
- password
- role

2. interviews

- id
- student_id
- interviewer_id
- interview_time
- status

3. coding_questions

- id
- title
- description
- difficulty

4. submissions

- id
- user_id
- question_id
- submitted_code
- verdict
- execution_time

5. feedback

- id
- interview_id
- AI_feedback
- rating

PHASE 1 — Authentication & Project Setup

Duration: 4–6 Days

Features:

- Signup/Login
- JWT Authentication
- Role-based access control
- Password encryption
- User management APIs

Workflow:

User Registers → Password Encrypted → Data Stored → Login → JWT Generated → Protected APIs Accessed

Skills Learned:

- Spring Security
- JWT
- API Security
- Database Integration

PHASE 2 — Interview Management System

Duration: 5–7 Days

Features:

- Create interview schedules
- Assign interviewers
- Track interview status
- Manage interview sessions

Important APIs:

- POST /interview/create
- GET /interview/all
- PUT /interview/status
- DELETE /interview/delete

Skills Learned:

- REST API Design
- Service Layer Architecture
- Entity Relationships

PHASE 3 — Coding Assessment Engine

Duration: 8–12 Days

Core Idea:

Students submit Java code.

Backend compiles and executes code safely.

Workflow:

Student Submits Code → Temporary Storage → Docker Container → Compilation → Test Cases → Result

Execution Strategy:

Phase 3A:

- Use Java ProcessBuilder

Phase 3B:

- Dockerized execution environment

Skills Learned:

- File Handling
- Multithreading
- Secure Code Execution
- Process Management

PHASE 4 — Real-Time Interview System

Duration: 5–7 Days

Features:

- Live interview room
- Real-time chat
- Live coding collaboration
- Timer management

Technologies:

- Spring WebSocket
- STOMP Protocol

Skills Learned:

- Real-time communication
- Event-driven backend systems
- Socket programming

PHASE 5 — AI Integration

Duration: 4–6 Days

Features:

- AI-generated coding feedback
- Resume suggestions
- Interview answer analysis
- Weakness detection

Integration:

Use OpenAI or Gemini APIs through RestTemplate or WebClient.

Skills Learned:

- Third-party API integration
- AI workflow design
- Asynchronous processing

PHASE 6 — Analytics Dashboard Backend

Duration: 4–5 Days

Features:

- Performance analytics
- Coding trends
- Leaderboards
- Weak topic analysis

Advanced Concepts:

- Aggregation queries
- Scheduled jobs
- Redis caching

Skills Learned:

- Backend optimization
- Data analytics
- Query optimization

PHASE 7 — Production-Level Enhancements

Duration: Ongoing

Advanced Features:

- Redis caching
- Kafka/RabbitMQ
- Swagger API Documentation
- Logging using SLF4J
- JUnit & Mockito Testing
- Docker Compose Deployment

Skills Learned:

- Production backend architecture

- Scalable system design
- DevOps basics

8. Resume & Placement Benefits

Project Title:

AI-Powered Mock Interview & Coding Assessment Platform

Resume Description:

Developed a scalable backend platform using Java Spring Boot featuring JWT authentication, real-time interview management, Docker-based secure code execution, AI-generated feedback, WebSocket communication, and analytics APIs.

Major Skills Demonstrated:

- Spring Boot
- REST APIs
- Security
- Docker
- WebSockets
- Redis
- AI Integration
- Backend Architecture

9. Final Notes

Final Advice:

1. Build phase by phase.
2. Push code regularly to GitHub.
3. Write proper README documentation.
4. Use Postman for API testing.
5. Focus heavily on backend quality and architecture.
6. Add Docker and WebSockets to stand out in placements.
7. Do not rush the project — quality matters more than quantity.

Expected Outcome:

This project can become your strongest placement project and can significantly improve your profile for backend development roles.